

MAJOR PROJECT REPORT ON

DEEPGUARD

Deepfake Image Detection Using CNN

SUBMITTED IN PARTIAL FULFILLMENT FOR AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE ENGINEERING

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Dec, 2025

Abstract

DeepGuard is an advanced machine learning project designed to address the growing threat posed by deepfake imagery and video manipulation. In recent years, advancements in artificial intelligence—particularly Generative Adversarial Networks (GANs)—have enabled the rapid creation of hyper-realistic fake visuals that are increasingly indistinguishable from authentic media. These manipulated artifacts pose serious risks in areas such as cybersecurity, political misinformation, identity fraud, and digital forensics. DeepGuard tackles this problem by implementing a carefully designed Convolutional Neural Network (CNN) model trained on a high-quality Kaggle real-vs-fake dataset. The project uses images standardized to 256×256 resolution and involves two independently trained models achieving accuracies of **96%** and **92%**, both maintaining losses under **0.15**, demonstrating strong generalization and stability.

To ensure accessibility and practical usability, DeepGuard is deployed through an interactive Streamlit dashboard capable of processing both images and videos. Users can upload media, preview it, and receive real-time predictions along with confidence scores that estimate the authenticity of the input. By combining deep learning, intuitive interface design, and robust evaluation techniques, DeepGuard offers a reliable and scalable solution for detecting deepfake media, setting the foundation for future expansions into real-time detection, multimodal analysis, and forensic-level authentication.

Declaration

I hereby certify that the work which is presented in the report entitled “Deepguard”. In fulfilment of the requirement for the award of the degree of bachelor of technology and submitted to the department of Computer Science Engineering of Sardar Beant Singh state university Gurdaspur. The matter presented in this dissertation has submitted by us for the award of this degree.

I further declare that in case of any violation of an intellectual property right or copyright we as the candidates would be fully responsible for the same. My department and university should not be held for full or partial violation of copyright if found at any stage of my degree.

Acknowledgement

We cannot achieve anything worthwhile in the field of technical education until or unless the theoretical education acquired in the classroom is effectively wedded to its practical approach that is taking place in the modern industries and research institute. Although an engineer can only be successful through hard work, but the contribution of his teachers and all those who have been helpful cannot be overlooked.

I am thankful to PROF. GURESH PAL SINGH for their exemplary guidance, monitoring and constant encouragement throughout the course of this training. Without their guidance this training would not have reached its final stage. The blessing, help and guidance given by them time to time shall carry me a long way in the journey of life.